

# Overview

This documentation describes the deployment of an Azure Application Gateway in front of Imaging v3.

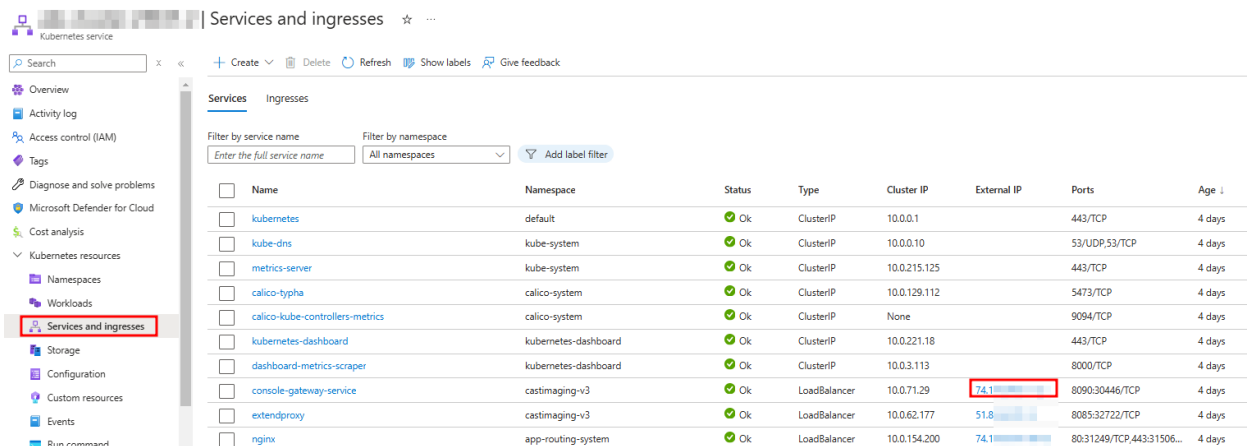
## Prerequisites

- You own a valid certificate in the PFX format with the associated passphrase.
- You have the required permissions on your Azure tenant to create Application Gateways.
- You have the required permissions to create a DNS record on the desired DNS zone which will point to your Application Gateway listener IP address.

## Procedure

### Retrieve console-gateway-service external IP

Retrieve the console-gateway-service pod external IP.

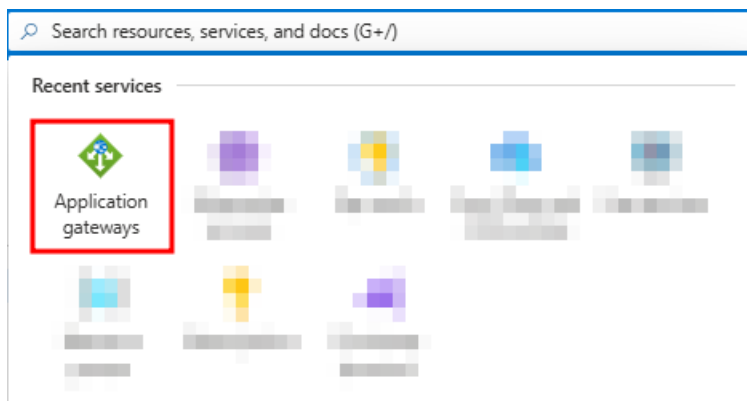


The screenshot shows the 'Services and ingresses' page in the Azure portal. The left sidebar contains a navigation menu with 'Services and ingresses' highlighted. The main area displays a table of services. The 'console-gateway-service' is highlighted, and its external IP address, 74.118.118.118, is shown in a red box.

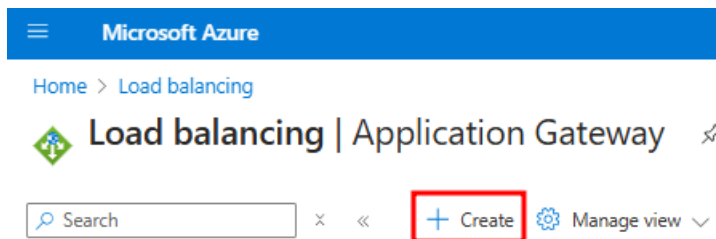
| Name                            | Namespace            | Status | Type         | Cluster IP   | External IP    | Ports                     | Age    |
|---------------------------------|----------------------|--------|--------------|--------------|----------------|---------------------------|--------|
| kubernetes                      | default              | Ok     | ClusterIP    | 10.0.0.1     |                | 443/TCP                   | 4 days |
| kube-dns                        | kube-system          | Ok     | ClusterIP    | 10.0.0.10    |                | 53/UDP,53/TCP             | 4 days |
| metrics-server                  | kube-system          | Ok     | ClusterIP    | 10.0.215.125 |                | 443/TCP                   | 4 days |
| calico-typha                    | calico-system        | Ok     | ClusterIP    | 10.0.129.112 |                | 5473/TCP                  | 4 days |
| calico-kube-controllers-metrics | calico-system        | Ok     | ClusterIP    | None         |                | 9094/TCP                  | 4 days |
| kubernetes-dashboard            | kubernetes-dashboard | Ok     | ClusterIP    | 10.0.221.18  |                | 443/TCP                   | 4 days |
| dashboard-metrics-scraper       | kubernetes-dashboard | Ok     | ClusterIP    | 10.0.3.113   |                | 8000/TCP                  | 4 days |
| console-gateway-service         | castimaging-v3       | Ok     | LoadBalancer | 10.0.71.29   | 74.118.118.118 | 8090:30446/TCP            | 4 days |
| extendproxy                     | castimaging-v3       | Ok     | LoadBalancer | 10.0.62.177  | 51.8.118.118   | 8085:32722/TCP            | 4 days |
| nginx                           | app-routing-system   | Ok     | LoadBalancer | 10.0.154.200 | 74.118.118.118 | 80:31249/TCP,443:31506... | 4 days |

## Create the Application Gateway

Search for the Azure Application Gateways service.



Create a new one.



Set the usual settings as desired. You will need to use a subnet dedicated to Application Gateways. Only use IPv4 IP address types as IPv6 is not yet supported on Imaging.

Microsoft Azure

Home > Load balancing | Application Gateway >

## Create application gateway ...

1 Basics

2 Frontends

3 Backends

4 Configuration

5 Tags

6 Review + create

An application gateway is a web traffic load balancer that enables you to manage traffic to your web application. [Learn about creating application gateway](#)

### Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription \*

Resource group \*

Create new

### Instance details

Application gateway name \*

Region \*

Tier

Availability zone \*

IP address type

HTTP2

my-app-gw

East US

Basic

Zones 1, 2, 3

☒ IPv4 only

☐ Dual stack (IPv4 & IPv6)

☐ Disabled

☒ Enabled

### Configure virtual network

Virtual network \*

Subnet \*

Create new

Manage subnet configuration

vnet-app-gw-

vnet-app-gw- subnet1 (10.0.1.0/24)

Previous

Next : Frontends >

Configure a new IP address or use an existing one.

Microsoft Azure

Home > Load balancing | Application Gateway >

Create application gateway ...

✓ Basics

**2 Frontends**

3 Backends

4 Configuration

5 Tags

6 Review + create

Traffic enters the application gateway via its frontend IP address(es). An application gateway can use a public IP address, private IP address, or one of each type. [↗](#)

Frontend IP address type ⓘ ☒ Public ☐ Private ☐ Both

Public IPv4 address \* 

(New) my-app-gw-pub-ip

[Add new](#)

Create a new backend pool. Use the console-gateway-service external IP you retrieved above.

## Add a backend pool. ×

Basic SKU supports a maximum of 5 backend pool targets

A backend pool is a collection of resources to which your application gateway can send traffic. A backend pool can contain virtual machines, virtual machines scale sets, IP addresses, domain names, or an App Service.

Name \* 

my-app-gw-backend-pool

Add backend pool without targets 

Yes **No**

Backend targets

1 item

| Target type                   | Target                                    |
|-------------------------------|-------------------------------------------|
| <div>IP address or FQDN</div> | <div><div></div> ✓ <div>⌵ ...</div></div> |
| <div>IP address or FQDN</div> | <div></div>                               |

Configure the listener. Choose the “Upload a certificate” option and upload a certificate which will be valid for the FQDN you plan to use for the Application Gateway.

Home > Load balancing | Application Gateway >

Create application gateway ...

✓ Basics ✓ Frontends ✓ Backends **Configuration** ⓘ Tags ⓘ Review + create

Create routing rules that link your frontend(s) and backend(s). You can also add more backend pools, add a second frontend IP configuration if you haven't already, or edit previous configurations. ⓘ

**Frontends**  
+ Add a frontend IP  
Public: (new) my-app-gw-pub-ip

**Routing rules**  
+ Add a routing rule

Previous Next: Tags >

### Add a routing rule

Basic SKU supports a maximum of 5 routing rules

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name \* my-app-gw-routing-rule ✓

Priority \* 100 ✓

\* Listener \* Backend targets

A listener "listens" on a specified port and IP address for traffic that uses a specified protocol. If the listener criteria are met, the application gateway will apply this routing rule to

Listener name \* my-app-gw-listener ✓

Frontend IP \* Public IPv4 ✓

Protocol HTTP HTTPS ✓

Port \* 443 ✓

**Https Settings**

Choose a certificate

☒ Upload a certificate ☐ Choose a certificate from Key Vault

Cert name \* my-app-gw-certificate ✓

PFK certificate file \* \*.pfx ✓

Password \* ..... ✓

Listener type Basic Multi site ✓

**Custom error pages**

Show customized error pages for different response codes generated by Application Gateway. This section lets you configure Listener-specific error pages. [Learn more](#) ⓘ

Please verify that the url(s) being added here is reachable from your application gateway using the [connection troubleshoot](#) tool to prevent any deployment error.

Bad Gateway - 502 Enter HTML file URL

Forbidden - 403 Enter HTML file URL

Add Cancel

In case the Application Gateway is shared with other applications, use this listener setting:

Listener type ⓘ

☐ Basic ☒ Multi site

Host type ⓘ

Single Multiple/Wildcard

Host name \* ⓘ

abc.xxxxxxxxxxx.com ✓

Configure the backend target with the following backend setting.

## Add Backend setting



Basic SKU supports a maximum of 5 routing rules

[← Discard changes and go back to routing rules](#)

Backend settings name \*  ✓

Backend protocol ☒ HTTP ☐ HTTPS

Backend port \*  ✓

### Additional settings

Cookie-based affinity ⓘ ☐ Enable ☒ Disable

Connection draining ⓘ ☐ Enable ☒ Disable

Request time-out (seconds) \* ⓘ

Override backend path ⓘ

### Host name

By default, the Application Gateway sends the same HTTP host header to the backend as it receives from the client. If your backend application/service requires a specific host value, you can override it using this setting.

☐ Yes ☒ No

Override with new host name

☐ Yes ☐ No

Create custom probes

Add a routing rule

×

Basic SKU supports a maximum of 5 routing rules

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name \*

my-app-gw-routing-rule

✓

Priority \* ⓘ

100

✓

\* Listener

\* Backend targets

Choose a backend pool to which this routing rule will send traffic. You will also need to specify a set of Backend settings that define the behavior of the routing rule. [↗](#)

Target type

☒ Backend pool ☐ Redirection

my-app-gw-backend-pool

▼

Backend target \* ⓘ

[Add new](#)

my-app-gw-backend-setting

▼

Backend settings \* ⓘ

[Add new](#)

**Path-based routing**

You can route traffic from this rule's listener to different backend targets based on the URL path of the request. You can also apply a different set of Backend settings based on the URL path. [↗](#)

| Path based rules                 |             |                      |              |
|----------------------------------|-------------|----------------------|--------------|
| Path                             | Target name | Backend setting name | Backend pool |
| No additional targets to display |             |                      |              |

Review the settings and click on the create button.

Microsoft Azure

Home > Load balancing | Application Gateway >

Create application gateway ...

Validation passed

Basics

Frontends

Backends

Configuration

Tags

6 Review + create

Basics

Subscription

sub\_dpt

Resource group

rg\_

Name

my-app-gw

Region

East US

Tier

Basic

Instance count

1

Availability zone

Zones 1, 2, 3

HTTP2

Enabled

Virtual network

vnet-app-gw-

Subnet

vnet-app-gw-

subnet1

Subnet address space

Frontends

Public IPv4 address name

my-app-gw-pub-ip

SKU

Standard

Assignment

Static

Availability zone

ZoneRedundant

Tags

None

Create

Previous

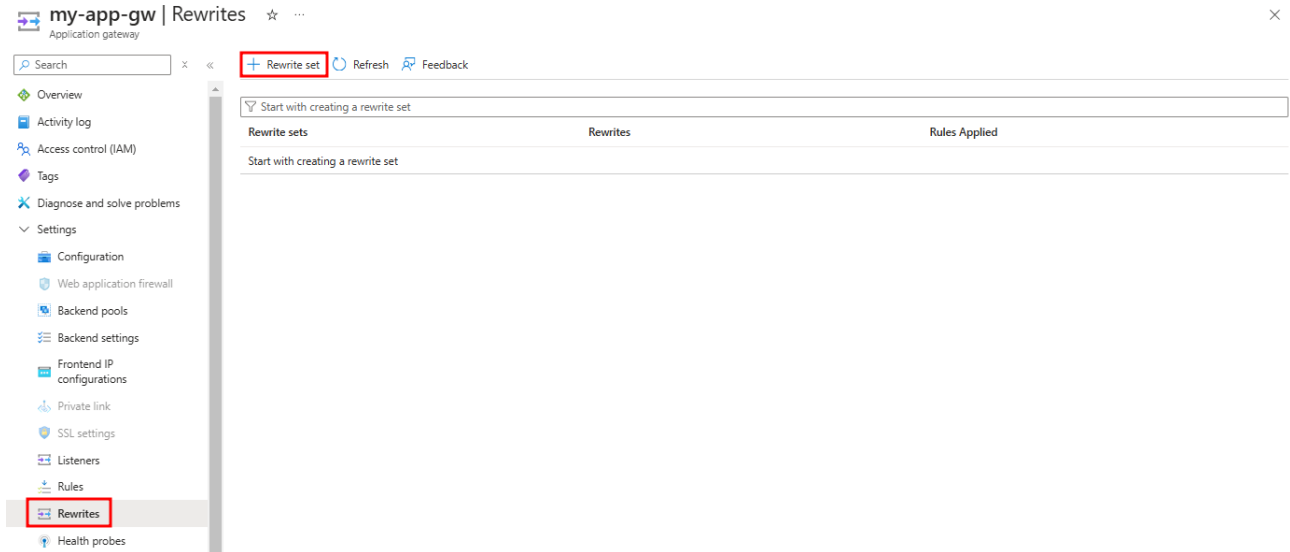
Next

Download a template for automation



Wait for your Application Gateway to be deployed, then create a new Rewrite set:

⇒ **Rewrites are only required if you are planning to enable SAML authentication in Imaging.**



Home > Load balancing | Application Gateway > my-app-gw | Rewrites >

## Create rewrite set ...

### 1 Name and Association 2 Rewrite rule configuration

To rewrite HTTP(S) headers, you need to create rewrite sets and associate them with routing rules. On this tab, you can provide the name to the rewrite set and associate it with the routing rules in your application gateway. On the next tab, you can configure the rewrite set by adding one or more rewrite rules to it. [Learn more about rewrite sets.](#)

Name \*

#### Associated routing rules

Select the routing rules to associate to this rewrite set. You can't select routing rules that already have an associated rewrite set.

| Routing rules   Paths                                      | Type       |
|------------------------------------------------------------|------------|
| <input checked="" type="checkbox"/> my-app-gw-routing-rule | Basic rule |

Create the following rewrite rules:

 Add rewrite rule

| Rewrite rule name  | Rule sequence | Condition                                                                                                                                                                                                                                                                                                         | Action          |             |               |                    |                                                                                                                                                                                                 |
|--------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|---------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                    |               |                                                                                                                                                                                                                                                                                                                   | Rewrite type    | Action type | Header name   | Common header      | Header value                                                                                                                                                                                    |
| cookie-rewrite     | 100           | <b>Type of variable to check:</b><br>HTTP header<br><br><b>Header type:</b><br>Response Header<br><br><b>Header name:</b><br>Common header<br><br><b>Common header:</b><br>Set-Cookie<br><br><b>Header value pattern to match:</b><br>.*<br><br><b>Operator:</b><br>equal (=)<br><br><b>Case-sensitive:</b><br>No | Response Header | Set         | Common header | Set-Cookie         | <b>Header value pattern to match:</b><br>.*<br><b>Operator:</b><br>equal (=)<br><b>Case-sensitive:</b><br>No<br><b>Set Header value to:</b><br>{http_resp_Set-Cookie};<br>SameSite=None; Secure |
| XForwardedHost     | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Set         | Common header | X-Forwarded-Host   | <your public Application Gateway FQDN><br>E.g. imaging.mydomain.com                                                                                                                             |
| XForwardedProtocol | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Set         | Common header | X-Forwarded-Proto  | https                                                                                                                                                                                           |
| XScheme            | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Set         | Custom header | X-Scheme           | https                                                                                                                                                                                           |
| XForwardedScheme   | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Set         | Custom header | X-Forwarded-Scheme | https                                                                                                                                                                                           |
| XRealIP            | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Delete      | Custom header | X-Real-IP          |                                                                                                                                                                                                 |
| XForwardedFor      | 100           |                                                                                                                                                                                                                                                                                                                   | Request Header  | Delete      | Custom header | X-Forwarded-For    |                                                                                                                                                                                                 |

Finally, you should see this set of rules:

#### Rewrite rules (Rule sequence)

cookie-rewrite (100)

XForwardedHost (100)

XForwardedProtocol (100)

XScheme (100)

XForwardedScheme (100)

XRealIP (100)

XForwardedFor (100)

cookie-rewrite rule details:

If

Type of variable to check \* ⓘ

HTTP header

Header type \*

Response Header

Header name \*

☒ Common header

☐ Custom header

Common header \*

Set-Cookie

Header value pattern to match ⓘ

.\*

Operator \*

equal (=)

Case-sensitive \* ⓘ

☐ Yes

☒ No

OK

Cancel

Then

Rewrite type \* ⓘ

Response Header

Action type \* ⓘ

Set

Header name \* ⓘ

☒ Common header

☐ Custom header

Common header \*

Set-Cookie

Header value pattern to match ⓘ

.\*

Operator

equal (=)

Case-sensitive ⓘ

☐ Yes

☒ No

Set Header value to \* ⓘ

{http\_resp\_Set-Cookie}; SameSite=None;...

OK

Cancel

## Create the Application Gateway DNS record

Make sure that the FQDN you plan to use for the Imaging deployment points to the Application Gateway public IP.

## Resolve issues

If you encounter issues when connecting through the Application Gateway, carefully review all the procedure steps. Also confirm that the Application Gateway backend is healthy. It should return a 302 as below.

The screenshot shows the 'Backend health' page for an Application Gateway named 'my-app-gw'. The left sidebar contains a navigation menu with items like 'Backend settings', 'Frontend IP configurations', 'Private link', 'SSL settings', 'Listeners', 'Rules', 'Rewrites', 'Health probes', 'Properties', 'Locks', 'Monitoring', 'Alerts', 'Metrics', 'Diagnostic settings', 'Logs', 'Insights', 'Backend health' (highlighted with a red box), and 'Connection troubleshoot'. The main content area is titled 'Backend health' and includes a description: 'By default, Azure Application Gateway probes backend servers to check their health and whether they're ready to serve requests. You can also create custom Health Probes to mention a specific hostname and path to be probed or a response code to be accepted as Healthy.' Below this, there are two summary boxes: 'All' showing '1 out of 1' with a red heart icon, and 'Healthy' showing '1 out of 1' with a green checkmark icon. A table below these boxes lists the backend health status for a single server in the pool.

| Server (backend pool) | Status  | Port (Backend setting)    | Protocol | Details                           | Action |
|-----------------------|---------|---------------------------|----------|-----------------------------------|--------|
| (my-app-gw-backend... | Healthy | 8090 (my-app-gw-backen... | Http     | Success: Received 302 status code |        |